

Capstone Project: Sales Analysis of the Christmas Season Sales Drop

Causes of the Christmas 2020, 2021 revenue and profit decline at the supermarket, and data metrics guiding next steps.

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Data Source: FP20 Analytics Challenge 12

Comparison Seasons: 2019, 2020 / 2020, 2021 / 2021, 2022

Source File: SQL Project Updated.sql

Executive Summary

Between the baseline season 2019, 2020 and the drop season 2020, 2021, Christmas revenue fell **-3.2% (-\$227,000)** and profit fell **-3.2% (-\$177,000)** in tandem, leaving profit margins virtually unchanged. The cause was not fewer shoppers: total transaction volume remained steady. Customers, specifically adult shoppers (18+) across all sales channels, bought smaller baskets and spent approximately **4% less per visit**. The 2021, 2022 season partially recovered, but Christmas Market and Online channels continued to contract even as In-Store broke past baseline levels.

Key Metrics Overview:

- **Revenue (Baseline Season → Drop Season):** -3.2%
- **Average Spend / Record:** -4.1%
- **Record Count / Transactions:** +1.0%
- **Profit Margin Change:** -0.04 percentage points

01. How to Read the SQL File in Plain Language

What is actually inside SQL Project Updated.sql and how to find the numbers you need:

1. **The file is a long script divided into clearly labeled sections:** Scrolling through the file, you will see large comments like SECTION 0, TASK 2,

and Question 4. Each section is an independent block, so you can run just that block in SQL Server Management Studio without running the whole file.

2. **"Section 0" is a data health check, not analytical findings:** Before analyzing, the file checks raw data for issues such as missing values, duplicate rows, whether profit equals revenue minus cost, and whether any numbers are negative. All checks passed cleanly (0 errors across all checks).
3. **Two "views" handle all data preparation:** `dbo.v_xmas_sales` is the clean, ready-to-use version of the raw sales table (retaining November, December, and January while labeling Christmas seasons). `dbo.v_season_roles` labels the baseline season (2019, 2020), drop season (2020, 2021), and recovery season (2021, 2022).
4. **"Questions 1-10" and "Task 2/4/5" hold the actual figures:** Questions 1-3 provide overall revenue trends. Task 2 breaks down the decline into 6 core metrics. Questions 4-10 and Tasks 4/5 repeat comparisons by channel, country, age group, product, time slot, and profitability.
5. **Terminology note: record_count:** You will see this term instead of "order count" or "customer count".
6. **How to read result tables:** Most tables repeat three columns: baseline season, drop season, and recovery season (or % change). Blank/NULL cells in the % change section mean "cannot calculate" (for example, if that segment did not exist yet).

02. Finding 1: Revenue Trends

Query Reference: Question 1, Question 2, Task 2 diagnostic query.

- **Baseline Season (2019, 2020):** \$7.18M
- **Drop Season (2020, 2021):** \$6.95M (-3.2%)
- **Recovery Season (2021, 2022):** \$7.09M (+1.9% vs drop season, still -1.3% vs baseline)

Total Christmas revenue fell from **\$7.18M to \$6.95M** between the baseline season and the drop season, a loss of about **\$227,000**. The 2021, 2022 season recovered **\$135,000** of that, but remained approximately **\$92,000** below baseline.

The drop stemmed from basket size, not foot traffic

Task 2 breaks down revenue using the formula record_count x average value per record:

Season	Record Count	Avg Value / Record	Units / Record	Revenue / Unit
Baseline Season	9.924	\$723,41	4,30	\$168,15
Drop Season	10.023 (+1,0%)	\$693,62 (-4,1%)	4,14 (-3,7%)	\$167,44 (-0,4%)
Recovery Season	10.034 (+0,1%)	\$706,25 (+1,8%)	4,22 (+1,9%)	\$167,23 (-0,1%)

Table Explanation: Record count increased slightly during the decline (more rows, not fewer), proving customers did not disappear. The declining factor was average value per record (-4.1%), which stemmed almost entirely from fewer units per basket (-3.7%) rather than cheaper products (revenue per unit was nearly unchanged at -0.4%). Shopping trips occurred at a similar frequency, but consumers purchased fewer items during each checkout.

O3. Finding 2: Profitability

Query Reference: Task 5 (5a overall trends, 5b loss by category, 5c loss by channel).

Task 5a: Profit & Profit Margin by Season

Season	Revenue	Cost	Profit	Profit Margin	Cost / Unit
Baseline Season	\$7,179M	\$1,656M	\$5,523M	76,93%	\$38,79
Drop Season	\$6.952M	\$1,606M	\$5.346M (-3.2%)	76,89%	\$38,69
Recovery Season	\$7,087M	\$1,637M	\$5,449M (+1,9%)	76,89%	\$38,64

Key Finding: Profit margins shifted by only 0.04 percentage points across all three seasons, and unit cost remained essentially identical. Profit dropped almost exactly in line with revenue (-3.2% vs -3.2%). This was a decline in sales volume and basket value, not a cost or margin issue.

Concentration of Profit Loss

Segment	Baseline Profit	Drop Season Profit	Profit Drop	% Change
Clothing (Category)	\$688.687	\$572.444	-\$116,242	-16,9%
Wearable Tech (Category)	\$220.099	\$162.204	-\$57,895	-26,3%
In-Store (Channel)	\$3.543.613	\$3.391.657	-\$151,956	-4,3%

Note: The "Cost" column in this dataset likely reflects COGS (cost of goods sold) and may not include staffing, delivery, rent, or marketing overhead. Margins shown here represent gross margins.

04. Finding 3: Customer Shopping Behavior

Query Reference: Question 7 (age/gender/payment), Task 4a (channel x age group).

The decline concentrated among adult shoppers (18+), appearing consistently across every channel they used.

Question 7a: Revenue by Age Group, Drop Season vs. Baseline Season

Age Group	Baseline Revenue	Drop Season Revenue	Revenue Drop	% Change	Avg Value / Record Change
Over 18 years old	\$2,323M	\$1,970M	-\$352,695	-15,2%	-11,7%
12 to 17 years old	\$2,142M	\$2,137M	-\$4,683	-0,2%	-5,7%
1 to 11 years old	\$2,714M	\$2,845M	+\$130,363	+4,8%	+3,5%

Adult shoppers (18+) dropped **\$352,695** in revenue, exceeding the company's total net loss (**\$227,000**), while spending in the children's category (1-11 years old) rose **+4.8%**.

Adults Cut Spending Across All Channels (Task 4a)

Channel	Baseline Revenue	Drop Season Revenue	Revenue Drop	% Change
In-Store	\$1,309M	\$1,129M	-\$179,664	-13,7%
Christmas Market	\$0,521M	\$0,420M	-\$101,389	-19,5%
Online	\$0,493M	\$0,422M	-\$71,643	-14,5%

Payment Method: Card Spending Dropped, Cash Up Slightly
(Question 7c)

Payment Method	Revenue Drop	% Change
PayPal	-\$112,095	-10,6%
Debit Card	-\$86,995	-10,8%
Credit Card	-\$84,923	-4,6%
Other	-\$6,607	-1,0%
Cash	+\$63,605	+2,3%

05. Finding 4: Sales Channel Performance

Query Reference: Question 4.

Question 4: Revenue by Channel Across All Three Seasons

Channel	Baseline Season	Drop Season	Recovery Season	Drop vs. Baseline	Recovery vs. Drop
In-Store	\$4,595M	\$4,399M	\$4,764M	-4,3%	+8,3%
Christmas Market	\$1,293M	\$1,230M	\$1,188M	-4,8%	-3,4%
Online	\$1,292M	\$1,323M	\$1,134M	+2,4%	-14,2%

- **In-Store:** Dipped slightly (-4.3%) during the drop season, then recovered fully past baseline (+8.3%).

- **Christmas Market:** Declined for two consecutive seasons (-4.8%, then -3.4%), reflecting a longer-term structural weakening.
- **Online:** Grew during the drop season (+2.4%), but then fell sharply in the recovery season (-14.2%).

06. Additional Context: Geography

Query Reference: Question 6 (6a country, 6b city, 6c recovery).

Question 6a: Largest Revenue Drops by Country, Drop Season vs. Baseline Season

Country	Revenue Drop	% Change
United Kingdom	-\$140,757	-14,7%
Poland	-\$96,951	-18,1%
Germany	-\$73,926	-7,5%
France	+\$78,331	+14,5%
Belgium	+\$56,306	+9,9%

Losses concentrated heavily in the **UK, Poland, and Germany**, while **France, Belgium, and Portugal** registered growth.

07. Summary of Findings & Next Steps for Team Analysis

This report investigated the Christmas sales performance drop across four main pillars: **Revenue Trends, Profitability, Customer Purchasing Behavior**, and **Channel Performance**. The insights below summarize why the decline occurred and outline how team members can build upon these findings to develop deeper analyses and targeted solutions.

1. Summary of Why Christmas Sales Performance Declined
Revenue & Profitability Overview

Metric	Baseline (2019/2020)	Drop Season (2020/2021)	Change
Total Revenue	\$7.18M	\$6.95M	-\$227,000 (-3.2%)

Metric	Baseline (2019/2020)	Drop Season (2020/2021)	Change
Gross Profit	\$5.52M	\$5.35M	-\$177,000 (-3.2%)
Profit Margin	76.93%	76.89%	-0.04%
Transaction Records	9,924	10,023	+1.0%
Average Spend / Record	\$723.41	\$693.62	-4.1%
Units / Record	4.30	4.14	-3.7%
Revenue / Unit	\$168.15	\$167.44	-0.4%
Segment Breakdown of Revenue Changes			
Category Dimension	Segment	Revenue Change	% Change
Customer Age Group	Adults (18+)	-\$352,695	-15.2%
Customer Age Group	Teens (12-17)	-\$4,683	-0.2%
Customer Age Group	Children (1-11)	+\$130,363	+4.8%
Product Category	Clothing (Profit)	-\$116,242	-16.9%
Product Category	Wearable Tech (Profit)	-\$57,895	-26.3%
Sales Channel	In-Store	-\$196,000	-4.3%
Sales Channel	Christmas Market	-\$63,000	-4.8%
Sales Channel	Online	+\$31,000	+2.4%
Geographic Region	United Kingdom	-\$140,757	-14.7%
Geographic Region	Poland	-\$96,951	-18.1%
Geographic Region	Germany	-\$73,926	-7.5%
Geographic Region	France	+\$78,331	+14.5%
Geographic Region	Belgium	+\$56,306	+9.9%

2. Strategic Action Plan to Drive Revenue Recovery & Prevent Future Drops

To address the identified drop drivers, stabilize performance, and drive future revenue growth, the leadership team should implement the following targeted business strategies:

1. **Drive Adult Basket Size Expansion (Clothing & Wearable Tech):**
 - a. **Strategy:** Introduce tiered cross-category promotional bundles combining high-performing children's goods with adult apparel and tech accessories (e.g., spend \$100 on kids' gifts to receive 20% off adult clothing or tech items).
 - b. **Objective:** Reverse the 4.1% spend decline among adult shoppers by increasing units per checkout back above baseline levels.
2. **Restructure & Optimize Sales Channels:**
 - a. **Christmas Markets:** Re-evaluate stall locations and operational footprints in declining market sites, reallocating budget toward top-performing physical stores or high-traffic pop-ups.
 - b. **Online Channel:** Implement dynamic free-shipping thresholds (e.g., free shipping on orders over 5 units) and personalized online checkout recommendations to prevent post-baseline contraction and encourage basket building.
3. **Tailor Regional & Payment Incentives:**
 - a. **Regional Campaigns:** Deploy aggressive, localized holiday marketing and loyalty rewards in key underperforming markets (United Kingdom, Poland, and Germany) while replicating successful merchandising tactics from growth regions like France and Belgium.
4. **Payment Options:** Partner with digital wallet providers to offer holiday cashback or installment incentives for debit/credit card payments to offset card spending contractions.
 - a. **Focus:** Confirm whether the dataset's **Cost** column includes operational/overhead costs beyond COGS, and integrate **Order ID tracking** when available.

08. Data Quality & Assumptions

Section 0 Verification Results (Verified on fp20c12, 2026-09-15):

- **Missing Values:** 0 across all columns
- **Duplicate Rows:** 0 completely duplicate rows
- **Revenue = Cost + Profit:** Exact 100% match (0 discrepancies)
- **Negative Values:** 0 in revenue, quantity, cost, profit, unit price columns

- **Data Timeframe:** January 2018 to January 2022 (Full Nov, Dec, Jan coverage for focus seasons)

Open Assumption: Table `dbo.xmas_sales` lacks an order ID column, so `record_count` represents product line items rather than distinct shopping carts or transactions.

Built from SQL Project Updated.sql